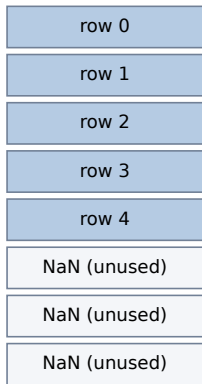


Python ColumnBuffer

one (capacity × columns) float64 array

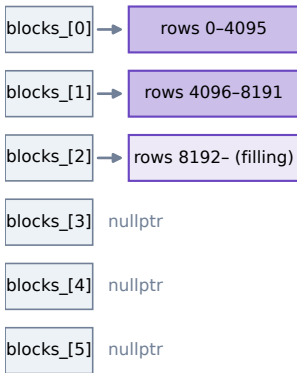


when full: allocate 2× rows,
copy the `_n` valid rows,
swap the reference.
A flush still holding the old
array keeps reading valid rows.

`_n = 5`: incremented AFTER
the row is written

C++ `fr::Stream`

fixed pointer table + 4096-row blocks



`n_`: `std::atomic<size_t>`

writer: fill the row, then
`n_.store(n+1, release)`

flush: `n_.load(acquire)`,
read rows below it

blocks never move or
get freed; at most
65 536 blocks
(≈268 M rows per stream)